



Smart Demand Forecasting and Shipment Delay Prediction

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Problem Statement



Retailers and logistics providers face operational inefficiencies from uncertain demand and fragile supply chains. Core challenges:

- **Overstocking** — inflated holding costs and obsolescence
- **Understocking** — missed sales, backorders, poor CSAT (Customer Satisfaction Score)
- **Shipment Delay** — business efficiency, customer satisfaction, and operational costs

These problems compound across scale and require predictive, data-driven interventions.



Proposed Solution

The solution integrates machine learning models with a Uvicorn-based web dashboard to deliver operational predictions and Delay Prediction.



Demand Forecasting

- Uses historical sales and shipment data to predict future demand.
- Helps maintain optimal stock levels and reduce overstock or stockouts.
- Improves cost efficiency and customer satisfaction through timely availability.



Delay Prediction

- Weather and carrier analytics forecast probable shipment delays.
- Improves supply chain efficiency and customer satisfaction.

Goals and Objectives

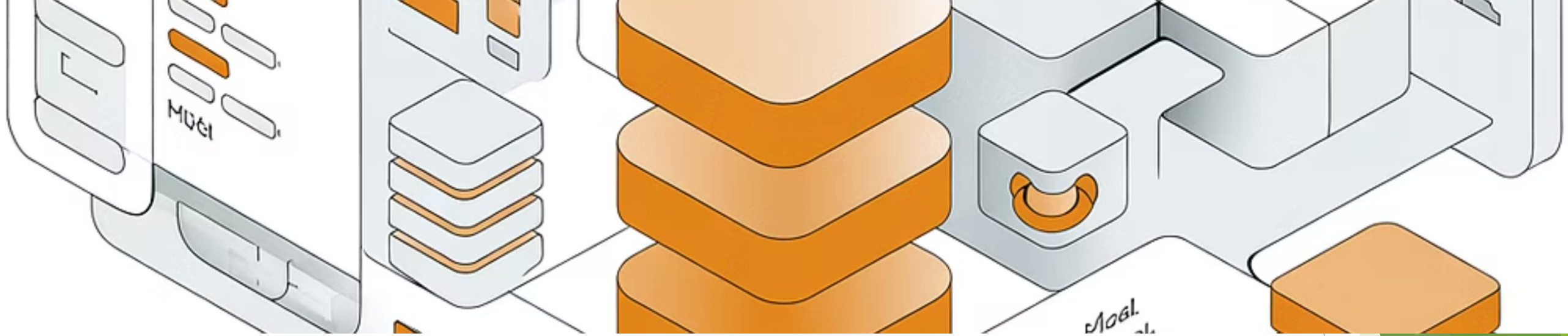
Main Goals

- Minimize overstock and stockouts (cost & service balance).
- Improve delivery reliability via predictive maintenance.
- Predict demand and shipment delays accurately.
- Minimize holding costs and inventory obsolescence.

Team Learning Objectives

- Build and deploy a Uvicorn dashboard integrating ML outputs
- Practice data preprocessing, feature engineering, and visualization
- Implement ML models for time-series and classification





Technology Used — Architecture Diagram

Key components



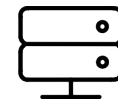
Frontend

HTML, CSS, JavaScript — interactive dashboards and visualizations



ML Libraries

Scikit-learn, Pandas, NumPy, Prophet, ARIMA — forecasting & anomaly detection



Backend

Python + Uvicorn for API endpoints, authentication, and serving models



Storage & fast APIs

JSON used as lightweight storage for user information.

Key Features —



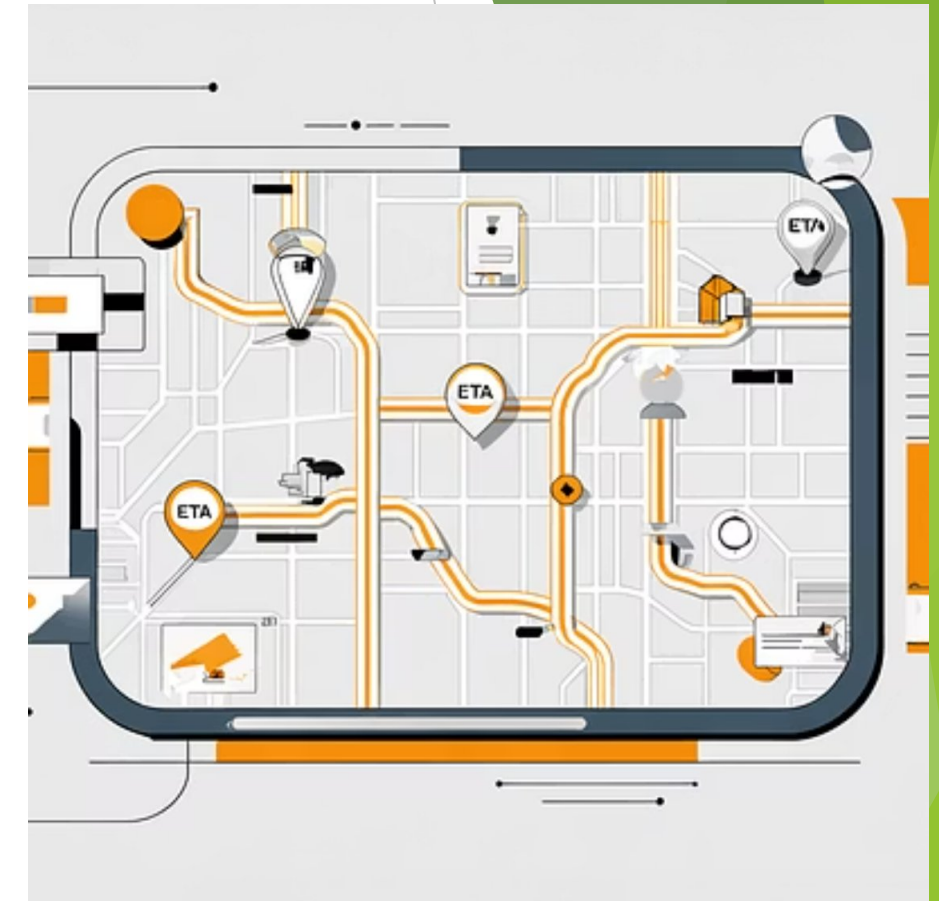
- Forecasting & Alerts

Week-ahead and month-ahead demand projections with confidence bands; automated reorder triggers.



- Operational feature

Instantly generate **7-day or 30-day demand forecasts** for any product.



- Interactive Visuals

Real-time charts, drill-downs, and CSV export for downstream analysis.

Future Prospects



1

Deep Learning

Integrate LSTM/Transformer architectures for longer-horizon, non-linear demand patterns.

2

Cloud Deployment

Move to cloud (GCP/AWS/Azure) for scalable, low-latency inference.

3

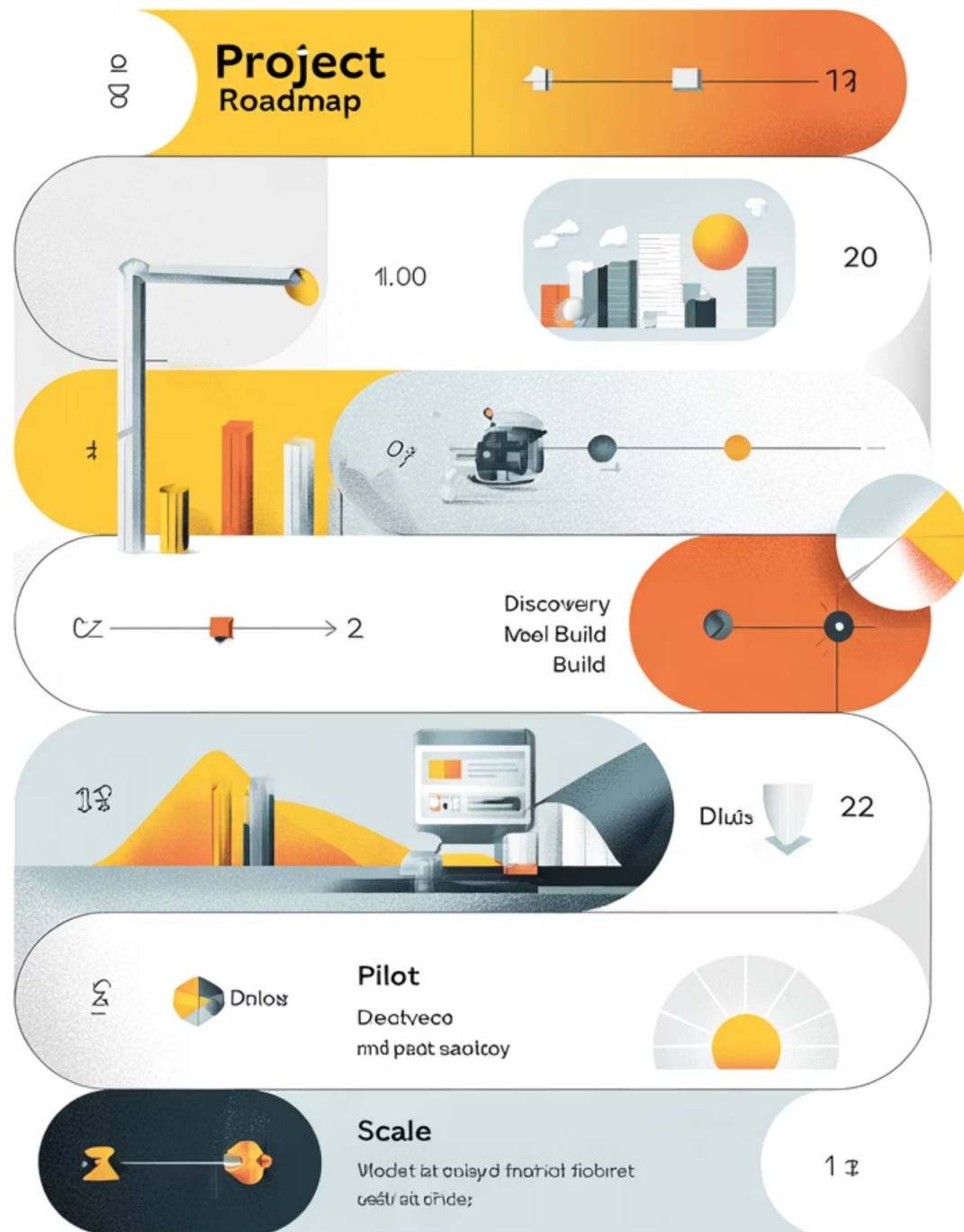
IoT Integration

Live vehicle telemetry and warehouse sensors for near-real-time decisioning.

4

Reinforcement Learning

Adaptive routing & warehouse robotics for continual policy improvement.



Implementation Roadmap

1. Discovery & Data Audit

Collect historical sales, telemetry, weather, and carrier data; identify gaps.

2. Model Development

Train/time-series and classification models; validate with backtesting.

3. Pilot & Integrate

- Evaluate accuracy, reliability, and business impact.
- Integrate the model with live inventory and logistics systems.

4. Production & Scale

Containerize, move to cloud, add monitoring, and roll out multi-warehouse support.

Thank You